

Orbits, Cycles, & Alternating Groups

Section 9

So $\sigma^{n+m}(a) = c$
and $a \sim c$. $\square$

Last time: Two-line notation for permutations, e.g.

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 7 & 4 & 1 & 5 & 2 \end{pmatrix}$$

Today: **cycles** of permutations.

Def: For $\sigma \in S_A$, partition A into cells given by $a \sim b$ if $\sigma^n(a) = b$ for some $n \in \mathbb{Z}$. These cells are called **orbits**.

Check: $\sim$ is equivalence relation.

- ① Reflexive: want $a \sim a$ for all $a \in A$. This is true since $\sigma^0(a) = e(a) = a$.
- ② Symmetric: $\nexists a \sim b$. Then $\exists n$ with $\sigma^n(a) = b$. Then $a = \sigma^{-n}(b) = a$, so $b \sim a$.
- ③ Transitive: $\nexists a \sim b$ and $b \sim c$. $\exists n, m$ with $\sigma^n(a) = b$ and $\sigma^m(b) = c$.

Ex:

$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 7 & 4 & 1 & 5 & 2 \end{pmatrix}$ has orbits $\{1, 3, 6\}$, $\{2, 8\}$, $\{4, 5, 7\}$

$$1 \xrightarrow{\sigma} 3 \xrightarrow{\sigma} 6 \xrightarrow{\sigma} 1$$

$$2 \xrightarrow{\sigma} 8 \xrightarrow{\sigma} 2$$

$$4 \xrightarrow{\sigma} 7 \xrightarrow{\sigma} 5 \xrightarrow{\sigma} 4$$

Cycle decomposition

$$\sigma = (1 \ 3 \ 6)(2 \ 8)(4 \ 7 \ 5)$$

$$\begin{aligned} (1 \ 3 \ 6) &= (3 \ 6 \ 1) \\ &= (6 \ 1 \ 3) \\ &\neq (1 \ 6 \ 3) \end{aligned}$$

Good convention: write the smallest number in a given cycle first.

Cycles as permutations

$(a_1 a_2 \dots a_k)$ is the permutation μ such that

$$\mu(x) = \begin{cases} a_{i+1} \pmod{k} & , \text{ if } x = a_i \\ x & , \text{ if } x \notin \{a_1, \dots, a_k\} \end{cases}$$

In S_3 : $(1\ 3) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$

Multiplying cycles

$$(1\ 3) \cdot (1\ 4\ 2) = (1\ 4\ 2\ 3)$$

$$(1\ 4\ 2) \cdot (1\ 3) = (1\ 3\ 4\ 2)^*$$

$$(2\ 3\ 1\ 4)$$

Disjoint cycles Two cycles are **disjoint** if they don't contain any of the same elements. Disjoint cycles commute.

Thm: Every permutation σ of a finite set has a (unique) decomposition as a product of disjoint cycles.

up to commuting the cycles and up to multiple ways of writing a given cycle

Pf: let $B_1, \dots, B_r$ be the orbits of σ , and

$$\mu_i(x) = \begin{cases} \sigma(x), & x \in B_i \\ x, & \text{otherwise} \end{cases}$$

Then $\sigma = \mu_1 \cdots \mu_r$.

Ex: $\sigma = e = \begin{pmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{pmatrix}$

$$e = (1)(2) \cdots (n).$$

Sometimes people do not write cycles for fixed points.

Ex: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 4 & 3 & 1 \end{pmatrix}$

$$\sigma = (1\ 6)(2\ 5\ 3)(4)$$

Ex: $(1\ 4\ 5\ 6)(2\ 1\ 5) = (1\ 6)(2\ 4\ 5)(3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 4 & 3 & 5 & 2 & 1 \end{pmatrix}$

$$(2\ 1\ 5)(1\ 4\ 5\ 6) =$$

exercise

Even & Odd permutations

Def: A cycle $(a_1 a_2)$ of two elements is a **transposition**.

Prop: Any permutation of a finite set is a product of (not necessarily disjoint!) transpositions. **NOT unique**

Pf: $(a_1 a_2 \dots a_k) = (a_1 a_k)(a_1 a_{k-1})(a_1 a_{k-2}) \dots (a_1 a_3)(a_1 a_2)$

$$a_1 \mapsto a_2$$

$$a_2 \mapsto a_1 \mapsto a_3$$

$$a_3 \mapsto a_1 \mapsto a_4$$

$$\vdots$$

$$a_k \mapsto a_1$$

Thus, we can write any $\pi \in S_n$ as a product of cycles, and then write each cycle as a product of transpositions. $\square$

Ex: $\sigma = (1\ 6)(2\ 5\ 3) = (1\ 6)(2\ 3)(2\ 5)$

Product of transpositions not unique. Nor is the number.

In S_3 : $\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = (1\ 2)$ $\sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (2\ 3)$

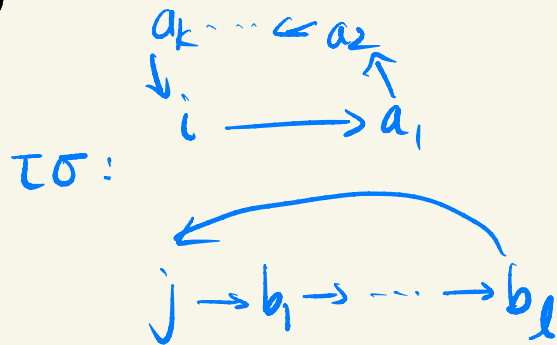
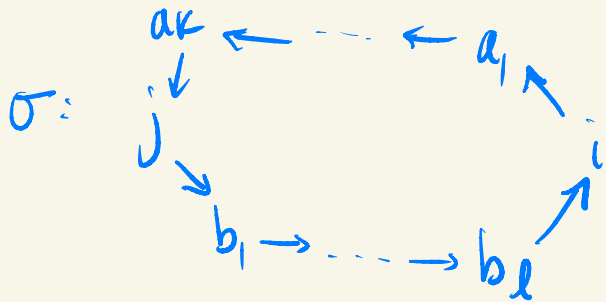
$$\sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2$$

$$(1\ 2)(2\ 3)(1\ 2) = (1\ 3)$$

Thm: No permutation in S_n can be expressed both as a product of an even number of transpositions and as an odd number.

Pf: Let $\sigma \in S_n$ and $\tau = (i\ j)$. We claim the number of cycles of $\tau\sigma$ and of σ differ by one.

Case 1: i, j are in the same cycle of σ .



Case 2: If i and j are in different cycles, then these cycles get merged in τ_0 . Exercise.

$$\overset{S_n}{\downarrow} e = (1)(2) \dots (n)$$

To finish the proof:

$$\sigma = \tau_1 \dots \tau_k$$

↖ transpositions

$$\# \text{ of cycles of } \sigma \equiv n - k \pmod{2}$$

$$\text{So if also have } \sigma = s_1 \dots s_{k'}$$

↖ transpositions

$$\text{Then } n - k \equiv n - k' \pmod{2}, \text{ so } k \equiv k' \pmod{2}. \quad \square$$

Def: A permutation is **even** or **odd** according to whether it can be expressed as a product of an even or odd # of transpositions.

Ex:

- e is **even** $e = (12)(12)$
- $(1\ 3\ 2)$ is $(1\ 3\ 2) = (1\ 2)(1\ 3)$ **even**
- $(12)(13)$ is **even**
- $(12)(132)$ is **odd**

The Alternating Groups

Let $A_n \subset S_n$ be the set of even permutations.

Thm: A_n is a subgroup of S_n and has order $n!/2$ for $n \geq 2$.

Pf:

$$t_1 \dots t_{2k}$$

① closure: $\pi, \sigma \in A_n$, then $\pi\sigma = t_1 \dots t_{2k} t'_1 \dots t'_{2l}$
this is an even # of transpositions.
so $\pi\sigma \in A_n$.

$$\begin{array}{l} A_n \xleftrightarrow{b_1} S_n \setminus A_n \text{ Thus } |A_n| = \frac{n!}{2} \\ \sigma \mapsto (12)\sigma \end{array}$$

② $e \in A_n$: since e is even

③ Inverse: $\pi \in A_n$. Then $\pi = t_1 \dots t_{2k}$ so $\pi^{-1} = t_{2k}^{-1} \dots t_1^{-1}$
so $\pi^{-1} \in A_n$ $= t_{2k} \dots t_1$ $\square$

A_n is called the **alternating group on n letters**. These groups are very important.

- We will see that for $n \geq 5$ they are **simple** groups.
- The structure of A_5 can be used to show the insolubility of quintic polynomials by radicals
 - **Galois theory**